

software details are available at <https://sourceforge.net/projects/iiot-gateway>. The Bufferstack.IO IIoT Gateway has a home at <http://bufferstack.io/>.

OEMs and systems integrators

Regarding the business model for this gateway, Joshi says that business in this sector revolves around two key components — OEMs and systems integrators (SI). The latter are experts appointed by OEMs to provide sales, service and support to solutions created using OEM hardware and software technology.

Joshi says, “The OEM and systems integrator are targeted by using the Platform as a Service (PaaS) and Software as a Service (SaaS), whereby the OEM uses the IIoT Gateway platform for adding remote control, monitoring, and other smart features and the SI designs applications as per client requirements. Apart from these two elements, freelancers and tech enthusiasts can tweak and use this OS for their own requirements.”

Joshi adds that on a personal level, designing his own OS and development tools made him a

confident and better professional than when he was using closed source software. It allowed him to enhance his programming problem-solving skills and resulted in an overall broader view towards software engineering. On a professional level, it gave him an opportunity to interact with leading developers and industry experts.

Joshi informs that he is currently in talks with freelancers and independent service vendors from the industrial automation field to test this gateway with industrial hardware that is used in factories for PLCs (programmable logic controllers) and VFDs (variable frequency drives). There are plans to hold system demonstrations, seminars and training for engineering students as well as industrial users after the lockdown is over, across Maharashtra and India. The sample application code has been released online on GitHub at https://github.com/hj91/iiot_gateway_python_sample_code.

Joshi is also working on other upcoming open source projects. One of these is the Super Container OS — a live Linux OS that makes use of Linux containers technology for

quick installation, deployment and distribution of cloud based applications. It is targeted towards the IT sector. After that, he has plans to develop computer vision applications using AI and machine learning tools for industrial automation.

Indianising global technology

These kinds of innovations will have a big impact on India, Joshi believes. He says, “In a very simple manner, this gateway comes with low-cost installation and maintenance. There are no exorbitant licensing fees for terminals. Linux systems operate much faster and rarely crash. Being open source, they are developed constantly and therefore any bugs are fixed rapidly.”

“This product is one small step to Indianise global technology to suit Indian needs and Indian requirements along with providing a stable and secure means to operate, develop and deploy IoT, IIoT, and even IT enterprise-level applications,” he concludes. 

 By: Sreejani Bhattacharya

The author is a tech correspondent at OSFY.

OSFY Magazine Attractions During 2020-21

MONTH	THEME
March 2020	Mobile/Web App Development and Optimisation
April 2020	Cybersecurity, Open Source Firewall, Network Security and Monitoring
May 2020	AI, Deep Learning and Machine Learning
June 2020	DevOps Special
July 2020	Blockchain and Open Source
August 2020	Database Management and Optimisation
September 2020	Open Source and IoT
October 2020	Cloud Special: BigData, Hadoop, PaaS, SaaS, IaaS and Cloud
November 2020	Open Source on Windows
December 2020	Open Source Programming (Languages and Tools)
January 2021	Data Security, Storage and Backup
February 2021	Best in the World of Open Source (Tools and Services)